



DATA ANALYST II- SQL PORTFOLIO

Entry Level Apprenticeship

Data Analysis 2 Portfolio

Professional Background

As a Certified Scrum Master with over 10 years of professional experience with a background in leading high-performance teams of 8-20 people across technology, business, and design departments. Expert in prioritizing tasks and optimizing workflows for organizational impact. Team-oriented professional with practical experience overseeing business metrics to remove impediments and ensure profitable product delivery. I am currently working as a University Registrar where I create ad-hoc reports on a weekly basis. My goal to attend Entry Level's Data Analyst course is to become a data whiz who can create lasting visualizations that capture the stories of profitable businesses. This portfolio report will further my proficiency with Sequenced Query Languages. I look forward to furthering my skills with SQL and Data Analytics.

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Introduction

This portfolio is based on a hypothetical situation that I am a Data Analyst working for the charity, Education for ALL. The Head of Fundraising has asked me to present the data on donor insights and donation rates.

My objectives are to:

- Increase the number of donors in our database.
- Increase the donation frequency of our donors. Increase the value of donations in our database.

This report will present insights from the donation which aims to inform my fundraising strategy to increase donations for the following year. Data sets FOR_Donation_Data and FOR_Donor_Data were used to answer the business problem.

The following SQL commands were used to analyze data: JOIN, ORDER BY, WHERE, BETWEEN, AND, OR, SUM(), COUNT(), AVG(), GROUP BY, HAVING.

The right question(s) will help understand the opportunity found from the insights provided by the data sets using prepared visualizations in this report.

Root Cause Analysis

Charity organizations are faced with the challenge of increasing funding and must research ways to grow their donorship to include recurrent donors who are likely to increase their donations. To understand the problem, two data bases of Donors and Donations were analyzed. Queries were visualized of the data sets to answer the following questions to answer how Education For ALL can grow donations and increase funding by analyzing:

- How many donors are in the existing databases?
- What is the amount of their donations?
- What is the frequency of their donations?
- Who are the top 10 donors?
- Does the amount of donations depend on gender, job field, or university degree?

Analysis

Two relational databases were analyzed to provide insights to the business challenge: Donation_Data and Donor_Data2.

SQLite Database Management System and Tableau Public were used to organize and visualize data.

Donation Dataset includes such data:

- Id
- First name
- Last Name
- Email
- Gender
- Job field
- Donation
- State
- Shirt Size

Donor Dataset includes such data: -

- Id
- Donation frequency
- University
- Car
- Second language
- Favorite color
- Movie genre

Both data sets were imported into SQLite.

SELECT query was used to Run data from the database.

```
SELECT *  
FROM Donation_Data
```

```
SELECT *  
FROM Donor_Data2
```

SELECT COUNT () To find the total number of donors to the charity. 1000

```
SELECT COUNT (donation)  
FROM Donation_Data;
```

SUM () query was used to find the sum of donations to the charity organization. 249085

```
SELECT Sum (donation)
FROM Donation_Data;
```

Top 10 donors:

```
1 SELECT Donation gender, donation, state, donor.frequency, donor university
2 FROM donation_data
3 JOIN Donor_Data 2
4 ON Donation_Data.id = donor_data.id
5 WHERE donation_frequency = "weekly"
6 AND donation BETWEEN 400 AND 500
7 ORDER BY donation DESC
8 LIMIT 10;
```

Also, I found out how many males and females without university education donated more than \$400.

20 Women over \$400 donate on a monthly basis.

gender	job_field	donation	state	donation_frequency	university
Female	Human Resources	450	Ohio	Monthly	null
Female	Accounting	444	Michigan	Monthly	McAlindon
Female	Product Management	483	Connecticut	Monthly	Gorch
Female	Sales	451	Florida	Monthly	Olive
Female	Business Development	421	Florida	Monthly	Effnert
Female	Marketing	423	Tennessee	Monthly	Littlejohn
Female	Legal	483	Pennsylvania	Monthly	null
Female	Support	462	Pennsylvania	Monthly	null
Female	Engineering	491	New Mexico	Monthly	Blackborn

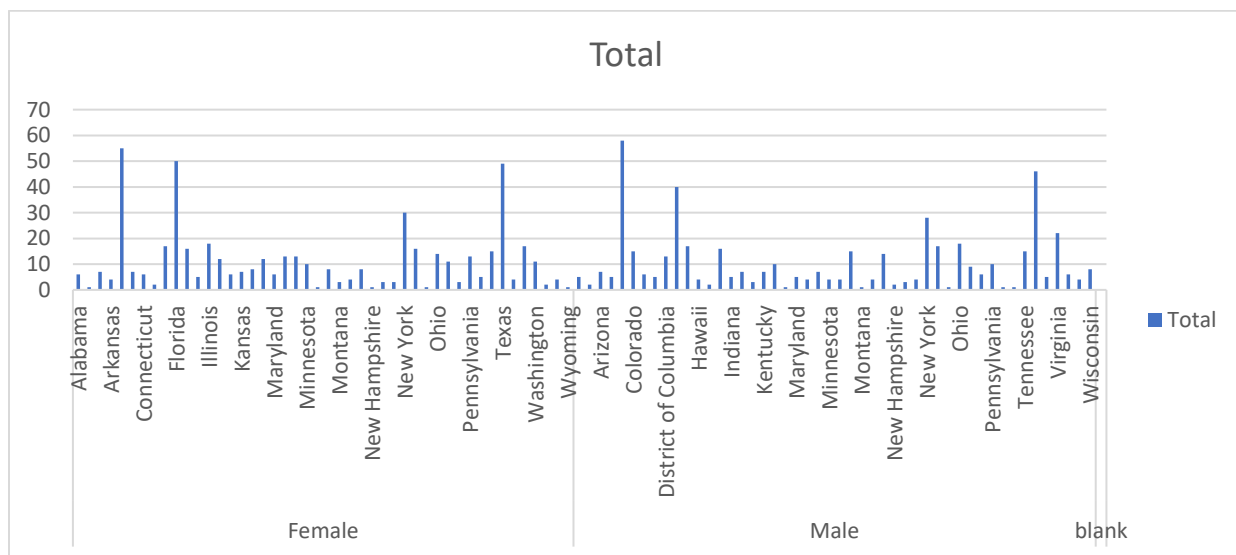
Female	Accounting	442	Virginia	Monthly	Laughrey
Female	Product Management	494	California	Monthly	Coates
Female	Business Development	451	Colorado	Monthly	Chrippes
Female	Sales	431	Florida	Monthly	Scullard
Female	Engineering	490	Kentucky	Monthly	Heinert
Female	Support	431	Kentucky	Monthly	Darton
Female	Services	407	Massachusetts	Monthly	null
Female	Human Resources	458	Colorado	Monthly	Maric
Female	Engineering	480	Arkansas	Monthly	null
Female	Support	426	Michigan	Monthly	null
Female	Training	442	District of Columbia	Monthly	Josuweit

18 Men donated over \$400 on a Monthly basis.

gender	job_field	donation	state	donation_frequency	university
Male	Support	458	California	Monthly	Mallall
Male	Human Resources	480	Tennessee	Monthly	null
Male	Human Resources	408	Ohio	Monthly	null
Male	Engineering	410	Michigan	Monthly	null
Male	Human Resources	441	Ohio	Monthly	Barnes
Male	Business Development	415	California	Monthly	Lennon
Male	Research and Development	427	California	Monthly	Statton
Male	Human Resources	411	Georgia	Monthly	Domenichelli
Male	Business Development	428	Missouri	Monthly	Ellacott
Male	Sales	411	California	Monthly	Ackeroyd
Male	Support	441	California	Monthly	null
Male	Research and Development	414	Tennessee	Monthly	Micklem

Male	Research and Development	409	New York	Monthly	Luppitt
Male	Business Development	442	Texas	Monthly	null
Male	Accounting	439	Florida	Monthly	Botte
Male	Support	418	Florida	Monthly	null
Male	Human Resources	494	California	Monthly	Baumber
Male	Training	437	Virginia	Monthly	Blackford

Tableau was used as a very powerful tool for data analysis and visualisation. Thus, the highest amounts of donations weekly coming from California, New York and Florida



Conclusion

Based on the findings, women are more likely than men to donate more than \$400 on a recurring basis than men were. The states with the highest donors include California, District of Columbia, New York, and Tennessee. Donors from Florida donated to the Education For ALL charity at a higher rate on a monthly level than donors from other states. Overall, donors from universities in all states donated at a higher frequency than donors with no collegiate affiliation. From this report we can see that donors with a college degree are more likely to donate more than \$40 on a recurring basis with the charity which is an opportunity for growth and additional funding. Specifically targeting graduates who live in states that have a higher GDP in the United States.

